

Mahesh Choudhary

 [LinkedIn](#) |  [email](#) |  [Mobile](#)

EDUCATION

Indian Institute of Technology Delhi

New Delhi , 110016

Bachelor of Technology in Electrical Engineering (8.577/10)

2020-2024

Relevant Coursework: Data structures and Algorithms, Analysis and Design of Algorithms, Computer Architecture, Machine Learning, Probability and Stochastic Processes, Embedded Systems ,Digital Electronics,Signals and Systems, Analog Electronic Circuits.

SCHOLASTIC ACHIEVEMENTS

JEE advanced: Secured rank 2320 among 1.5 lakh candidates appeared for examination (2020).

Merit scholarship: awarded by IIT Delhi for among top 7 percent of students in academic performance.

Competitive Programming: Recognised as Expert at codeforces ([mcl](#))

WORK EXPERIENCE

Digital Engineer Intern @Texas Instruments

May 2023 - July 2023

- Analysed working and verilog HDL codes of various components of UCD31xx(Digital Power Supply Controller) filter module.
- Designed model to use idle components of filter and developed proper ordering of computation to reduce best case required clock cycles from 8 to 6.

PROJECTS

Movie Recommendation System

Feb 2023 - April 2023

- Implemented movie recommendation system to give personalized movie suggestions to users based on their history and ratings given by them, and implemented update method to add new ratings in dataset.
- Computed most similar user to current using Jaccard similarity coefficient to suggest new unwatched movies.
- Used movieslens-100k dataset as initial data, maintained it with further updates in user-item-matrix.

Cryptocurrency DSCoin(Prof. Amitabha Bagchi)

Sept 2021 - Nov 2021

- Created functional cryptocurrency model, through utilisation and incorporation of various data structures.
- Implemented Authenticated lists and Merkle trees to ensure secure storage and swift transfer of data in blockchain.
- Devised a method to identify fraudulent transactions and reward miners for aid in decentralised funds exchange.

Academic Blockchain Document(Prof. Venkata Koppula)

Oct 2021

- Implemented BlockChain to store result of students' exams and to provide digital certificate as proof of their result.

Integration of DER and Storage devices in grid(Prof. Ankit Singhal)

Aug 2023 - Nov 2023

- Maintained generators, storage devices and loads in graph as nodes and interconnections as edges along with their states.
- Designed charge-discharge-transfer model to smoothen power curve exploiting storage device integration using greedy algorithms.
- Implemented methods to add new devices, interconnections, charge-discharge storages, and to transfer loads.

SKILLS

Languages: C++, C, Python, Java, MySQL, Verilog HDL, Latex.

Softwares and Libraries: Numpy, Matplotlib, Linux, Git, VS Code, Matlab.

POSITIONS OF RESPONSIBILITY

Mentor, Board for Student Welfare(2022-2023): Mentored five freshmen for stress-free transition in college.

Activity Head, Tryst'22 (Feb 2022 - April 2022: Introduced new events, handled event planning and scheduling, promoted events of the festival using social media platforms.